

Data Mining Project Report

Title

CFPB Consumer Complaints: Predicting Monetary Relief Outcomes from
Complaint-Time Information

Names

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Course Title

BA 360: Business Data Mining

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1 Dataset Description

1.1 General Context

The dataset comes from the U.S. Consumer Financial Protection Bureau (CFPB). It contains real consumer complaints about financial products and services (for example credit cards, loans, credit reporting, and debt collection). In business terms, this dataset helps identify where customer harm occurs and where monetary relief is more likely, which supports better risk monitoring and faster case prioritization.

1.2 Data Source

- Source: CFPB Consumer Complaint Database.
- Official page: <https://www.consumerfinance.gov/data-research/consumer-complaints/>
- Direct file used as full raw input in this project: `data/raw/complaints.csv`.
- Because the full file is very large, we created and used a memory-safe working sample for modeling: `data/processed/complaints_sample.csv`.

Local snapshot (as documented in the project README on 2026-04-20):

- Full raw dataset size: 14,636,145 rows and 18 columns.
- Uncompressed CSV size: about 8.0 GB.
- Compressed download size (`complaints.csv.zip`): about 1.7 GB.

1.3 Unit of Observation

One observation (one row) represents one consumer complaint submitted to the CFPB complaint system. Each row includes complaint metadata available at intake time (for example product, issue, submission channel, state, and dates), and in many cases a free-text complaint narrative.

For this project, we defined a binary target from resolution information:

- 1: complaint ended with monetary relief.
- 0: complaint ended without monetary relief.

To avoid leakage, post-resolution fields are not used as predictors at inference time.

1.4 Main Variables

Main variables used in EDA and modeling include:

- Text variable: `consumer_complaint_narrative`.
- Product taxonomy: `product` and `sub_product`.
- Problem taxonomy: `issue` and `sub_issue`.
- Submission and geography: `submitted_via` and `state`.
- Time variables: `date_received` (and derived temporal features).
- Target construction fields: `company_response_to_consumer` (used to derive the label, then excluded from predictor set for leakage-safe modeling).

Data handling note: because the full dataset in `data/raw/complaints.csv` is too large for efficient iterative experimentation on a local machine, we generated a representative sample

(data/processed/complaints_sample.csv) and used that sample throughout preprocessing and model development.

2 Problem Statement

2.1 Problem to Study

State the analytical problem clearly.

2.2 Objective

Define the objective of the analysis.

2.3 Research Questions

List the main research questions.

2.4 Target Variable

Explain the binary target design (monetary relief vs no monetary relief) and leakage-safe constraints.

3 Data Preparation

3.1 Missing Values

Explain handling strategy and justification.

3.2 Outliers and Quality Checks

Describe checks and decisions.

3.3 Feature Transformations

Detail text cleaning, vectorization, and any transformations.

3.4 Encoding and Scaling

Document categorical encoding and scaling choices.

3.5 Feature Selection and Leakage Control

List excluded post-resolution fields and explain why.

4 Methods and Analysis

4.1 Modeling Strategy

Explain train/test setup and model comparison approach.

4.2 Models Implemented

- Logistic Regression (TF-IDF)
- Naive Bayes (TF-IDF)
- KNN (SVD-reduced text)
- Random Forest (structured + text)
- MLP Neural Network
- Voting Classifier Ensemble

4.3 Why These Methods

Justify method choices for this problem.

5 Results and Interpretation

5.1 Model Comparison

Insert summary table from reports outputs.

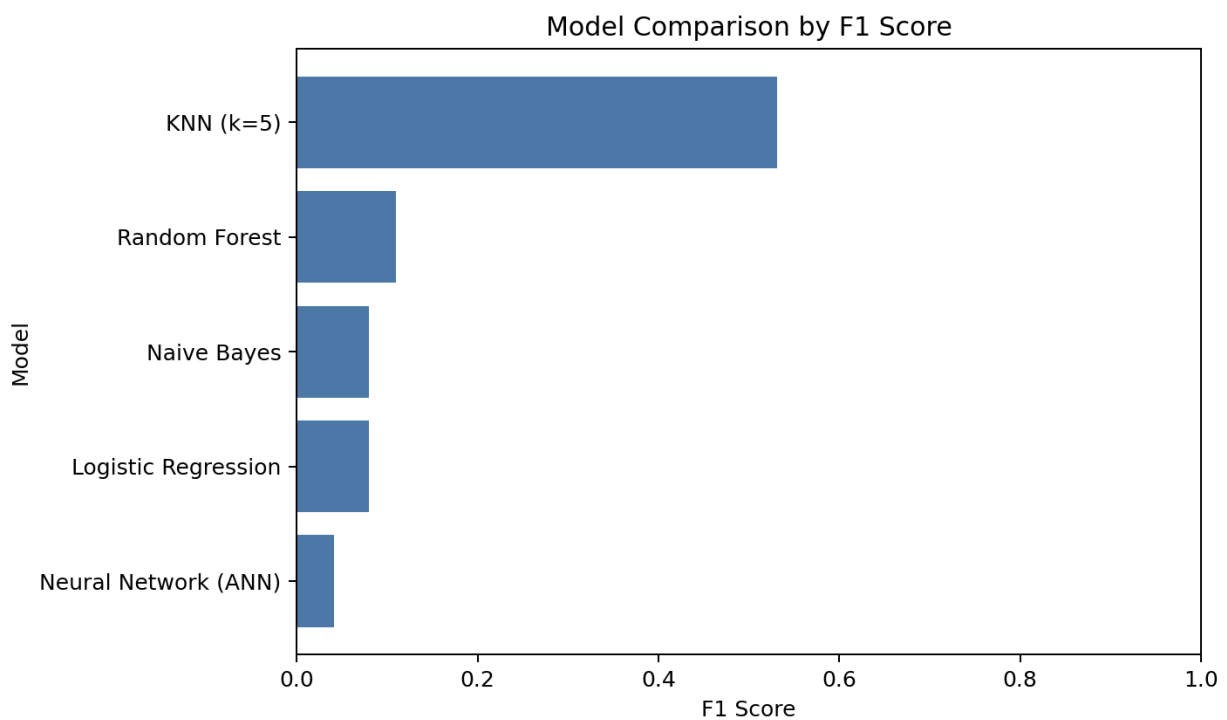


Figure 1: Model comparison table figure

5.2 Threshold Analysis

Discuss trade-offs and operational threshold recommendation.

5.3 Confusion Matrix Insights

Interpret false positives and false negatives in business terms.

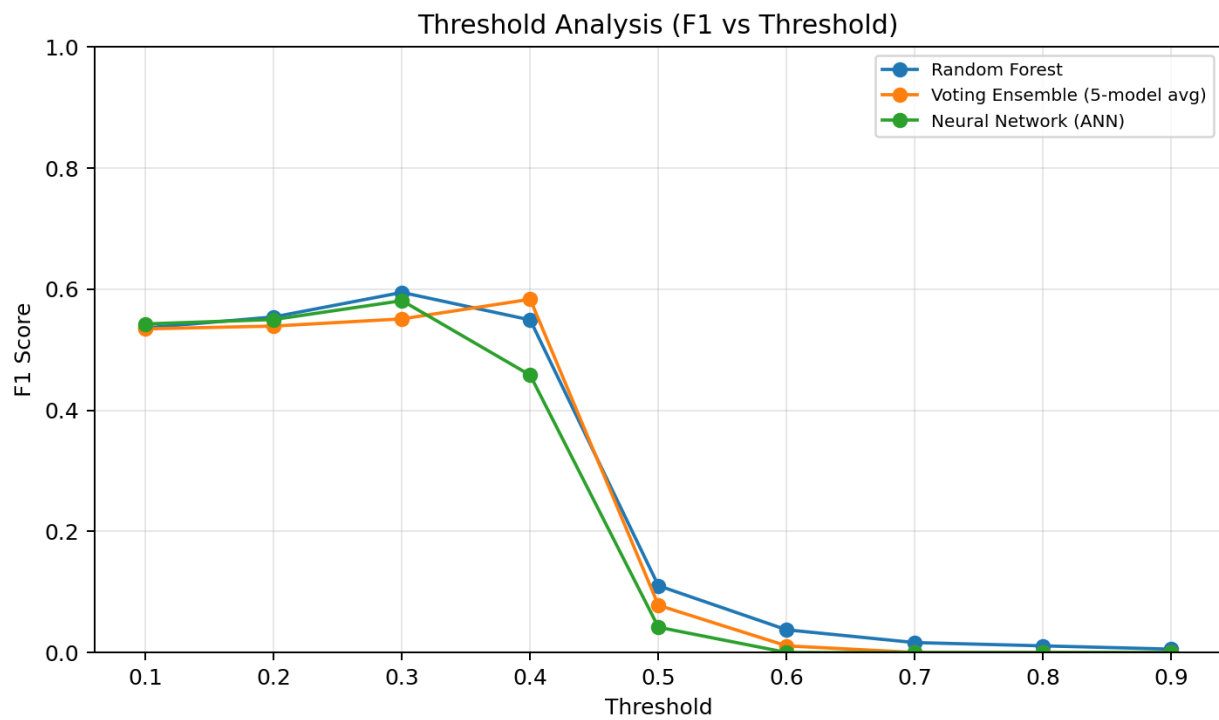


Figure 2: Threshold analysis

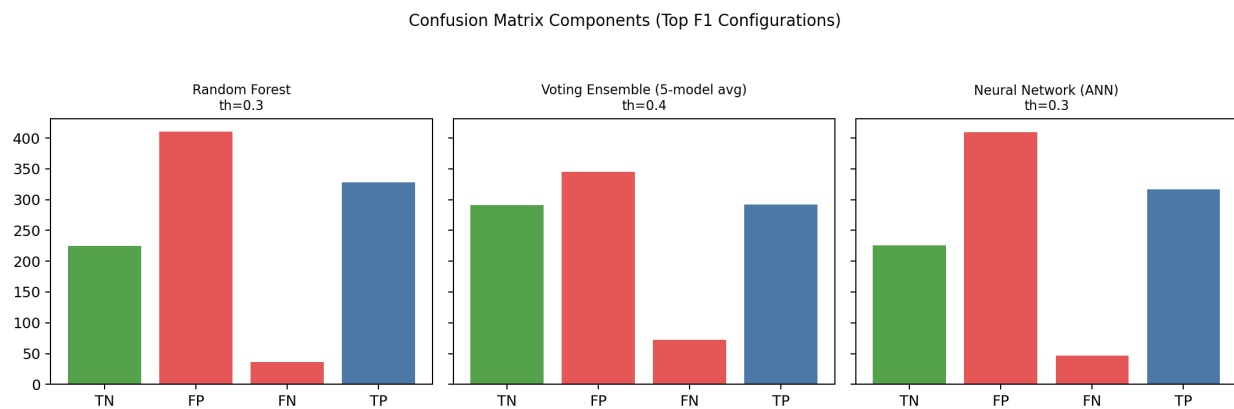


Figure 3: Confusion matrix

5.4 Key Findings

Provide clear, practical conclusions.

6 Conclusion

6.1 Objective Recap

Summarize initial objective.

6.2 Methods Recap

Summarize methods used.

6.3 Main Results

Summarize strongest findings and selected model.

6.4 Business Insights

Summarize actionable recommendations.

7 Appendix (Optional)

7.1 Evolution and Corrections

Reference the mistakes-to-improvements timeline.

7.2 Additional Tables

Add detailed model metrics if needed.